7 Prove or give a counterexample: If U is a nonempty subset of $\mathbf{R}^2$ such that U is closed under addition and under taking additive inverses (meaning $-u \in U$ whenever $u \in U$), then U is a subspace of $\mathbf{R}^2$.

Counterexample: Let $U = \{(x,y) \mid x, y \in \mathbb{Z}\}$.
Then U is closed under addition and taking additive inverses. Furthermore, 0 is in the set.
But, it is not closed under scalar multiplication.
Because $0.9(1,1) = (0.9, 0.9) \notin U$.
$\therefore U$ is not a subspace of $\mathbb{R}^2$.

11 Prove that the intersection of every collection of subspaces of V is a subspace of V.

Let V_{S_i} be a subspace of V.
Let V_n be an intersection of any n subspaces $V_{S_i}\big|_{i=1,\dots,n}$. Consider elements $x, y \in V_n$.
Since x, y are elements of V_n, they must be elements of $V_{S_1}, V_{S_2}, \dots, V_{S_n}$. i.e x and y are closed under addition and scalar multiplication. Also, since the additive identity exists in each of V_{S_i}, it exists in V_n.
$\therefore V_n$ is a subspace of V.

15 Suppose U is a subspace of V. What is $U + U$?

Then $U + U = \{u_1 + u_2 \mid u_1, u_2 \in U\}$
But $u_1 + u_2 \in U$ because U is closed under addition.
So $U + U = \{u \mid u \in U\} = U$
$\therefore U + U = U$.

16 Is the operation of addition on the subspaces of V commutative? In other words, if U and W are subspaces of V, is $U + W = W + U$?

$U + W = \{u + w \mid u \in U, w \in W\}$
and $W + U = \{w + u \mid w \in W, u \in U\}$
But $u + w = w + u$ because $u, w \in V$ and vectors in vector spaces are commutative.
$\therefore$ Yes, addition on the subspaces of V is commutative.

17 Is the operation of addition on the subspaces of V associative? In other words, if V_1, V_2, V_3 are subspaces of V, is

$$(V_1 + V_2) + V_3 = V_1 + (V_2 + V_3)?$$

$(V_1 + V_2) + V_3 = \{(v_1 + v_2) + v_3 \mid v_1 \in V_1, v_2 \in V_2, v_3 \in V_3\}$
and
$V_1 + (V_2 + V_3) = \{v_1 + (v_2 + v_3) \mid v_1 \in V_1, v_2 \in V_2, v_3 \in V_3\}$
But $(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3)$ because $v_1, v_2, v_3 \in V$ and vectors in V are associative.
$\therefore$ Yes.

20 Suppose

$$U = \{(x, x, y, y) \in \mathbf{F}^4 : x, y \in \mathbf{F}\}.$$

Find a subspace W of $\mathbf{F}^4$ such that $\mathbf{F}^4 = U \oplus W$.

$W = \{(0, a, 0, b) \in \mathbb{F}^4 : a, b \in \mathbb{F}\}$ works.
Verify: $(0,0,0,0)$ can ONLY be written when both $(0,0,0,0) \in U$ and $(0,0,0,0) \in W$.
Therefore, $U + W$ is a direct sum.
$U + W = \{x + 0, x + a, y + 0, y + b\}$
$= \{p, q, r, s\} = \mathbb{F}^4$.

23 Prove or give a counterexample: If V_1, V_2, U are subspaces of V such that

$$V = V_1 \oplus U \quad \text{and} \quad V = V_2 \oplus U,$$

then $V_1 = V_2$.

Hint: When trying to discover whether a conjecture in linear algebra is true or false, it is often useful to start by experimenting in $\mathbf{F}^2$.

$V = \mathbb{F}^2$
Counterexample: $V_1 = \{(x, 0) : x \in \mathbb{F}\}$
$V_2 = \{(0, y) : y \in \mathbb{F}\}$
$U = \{(a, a) : a \in \mathbb{F}\}$
Then $V = V_1 \oplus U$ because 1. Their intersection is only $(0,0)$.
2. $(x, y) = V$ is their sum.
and $V = V_2 \oplus U$ because 1. Their intersection is only $(0,0)$.
2. $(a, y + a) = V$ is their sum.
But $V_1 \neq V_2$. Disproved.

24 A function $f : \mathbf{R} \to \mathbf{R}$ is called *even* if

$$f(-x) = f(x)$$

for all $x \in \mathbf{R}$. A function $f : \mathbf{R} \to \mathbf{R}$ is called *odd* if

$$f(-x) = -f(x)$$

for all $x \in \mathbf{R}$. Let V_e denote the set of real-valued even functions on $\mathbf{R}$ and let V_o denote the set of real-valued odd functions on $\mathbf{R}$. Show that $\mathbf{R}^{\mathbf{R}} = V_e \oplus V_o$.

Consider a function $f : \mathbb{R} \to \mathbb{R}$ that is both even and odd.
i.e $f(-x) = f(x)$
and $f(-x) = -f(x)$
$+$
$f(-x) = 0 = f(x)$.
$\therefore$ The 0 function is the only odd and even function.
i.e $V_e \cap V_o = 0$

Consider an arbitrary function $f: \mathbb{R} \to \mathbb{R}$.
Let $f_{odd} \in V_o$, $f_{even} \in V_e$.

Case 1: f is even:

$$f = \underset{f_{odd}}{0} + f_{even}$$

Case 2: f is odd:

$$f = \underset{f_{even}}{0} + f_{odd}$$

Case 3: f is neither:

$$f = f_{odd} + f_{even}.$$

Let $f_{even} = \dfrac{f(x) + f(-x)}{2}$

and $f_{odd} = \dfrac{f(x) - f(-x)}{2}$

then $f_{odd} + f_{even} = f$.

We have proven that every $f: \mathbb{R} \to \mathbb{R}$ can be written as a sum of $f_{even} \in V_e$ and $f_{odd} \in V_o$.
Further, $V_e \cap V_o = 0$.
$\therefore \mathbb{R}^{\mathbb{R}} = V_e \oplus V_o$.